

To prove that the set of rational numbers $\mathbb{Q}$, equipped with the two binary operations of addition and multiplication, forms a field.

Soln: A set F with two binary operations $+$ and $\cdot$ is a field if the following hold:

1. $(F, +)$ is an abelian (commutative) group.

- a. closure under $+$,
- b. associativity of $+$,
- c. identity element 0
- d. additive inverses
- e. commutativity of $+$.

2. $(\mathbb{F} \setminus \{0\}, \cdot)$ is an abelian group.

a) closure under $\cdot$.

b) associativity of $\cdot$.

c) identity element 1.

d) inverse multiplicative for every non zero element

e) commutativity of $\cdot$.

③) Distributivity: $x \cdot (y + z) = x \cdot y + x \cdot z$

for all $x, y, z \in \mathbb{F}$

finally, $0 \neq 1$ must hold (so the two identities are distinct).

verification for $\mathbb{Q}$:

Every rational number can be written as $\frac{a}{b}$ with $a \in \mathbb{Z}, b \in \mathbb{Z} \setminus \{0\}$

1. $(\mathbb{Q}, +)$ is an abelian group.

• closure under addition:

if $x = \frac{a}{b}$ and $y = \frac{c}{d}$ then

$$x + y = \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

and $ad + bc$ and bd are integers with $bd \neq 0$. Thus $x + y \in \mathbb{Q}$.

Associativity: addition of rationals is associative because it follows from associativity of integer addition. For rationals x, y, z

$$(x + y) + z = x + (y + z).$$

Additive identity: 0 satisfies $x + 0 = x$ for every rational x .

Additive inverse: for $x = \frac{a}{b}$, the additive inverse is $-x = -\frac{a}{b}$, which is rational and satisfies $x + (-x) = 0$.

commutativity:

$$\frac{a}{b} + \frac{c}{d} = \frac{c}{d} + \frac{a}{b}$$

Hence $(\mathbb{Q}, +)$ is an abelian group.

2. $(\mathbb{Q} \setminus \{0\})$ is an abelian group

- closure under multiplication:

with $x = \frac{a}{b}, y = \frac{c}{d}$ $x \cdot y = \frac{ac}{bd}$

and ac, bd , are integers with $bd \neq 0$, so the product is in $\mathbb{Q}$. If neither x nor y is zero then $ac \neq 0$, so the product is non zero.

- * Associativity: multiplication of rationals is associative.

- * multiplicative identity: 1 satisfies $1 \cdot x = x$ for all $x \in \mathbb{Q}$

- * multiplicative inverse: for a non-zero rational $x = \frac{a}{b}$ with $a \neq 0$, the inverse is $\frac{b}{a}$ (an element of $\mathbb{Q}$)

and $\frac{a}{b} \cdot \frac{b}{a} = 1$

• commutativity: $\frac{a}{b} \cdot \frac{c}{d} = \frac{c}{d} \cdot \frac{a}{b}$ because
integer multiplication is commutative.
Thus $(\mathbb{Q}, \cdot)$ is an abelian group.

3. Distributivity: for rationals $x = \frac{a}{b}$,

$$y = \frac{c}{d}, z = \frac{e}{f}; \quad x(y+z) = \frac{a}{b} \left(\frac{c}{d} + \frac{e}{f} \right)$$

$$= \frac{a}{b} \cdot \frac{cf+de}{df}$$

$$= \frac{acf}{bdf} + \frac{ade}{bdf}$$

$$= \frac{a}{b} \cdot \frac{c}{d} + \frac{a}{b} \cdot \frac{e}{f}$$

$$= xy + xz.$$

4. $0 \neq 1$

In $\mathbb{Q}$, 0 is $0/1$ and 1 is $1/1$. These
are different rationals, so

$0 \neq 1$. This prevents the degenerate one-element ring.

All field axioms hold for $Q: (Q, +)$ is a abelian group, $(Q \setminus \{0\}, \cdot)$ is an abelian group multiplication distributes over addition, and $1 \neq 0$. There some Q with usual addition and multiplication is a field.